Risk of Progression from AK to SCC

In a review of five clinical studies conducted between 1988 and 1998, Glogau reported that the annual risk of progression from actinic keratosis (AK) to squamous cell carcinoma (SCC) for individual lesions ranged from 0.025 percent to 16 percent. Extrapolation from these studies suggests an average progression risk of approximately 8 percent per year.

Berhane et al. observed that AKs may become clinically tender and inflamed prior to transformation into SCC. Histopathologic examination of such inflamed lesions revealed the presence of invasive SCC in 50 percent of cases. Thus, pain and inflammation in an AK may serve as clinical markers of malignant progression.

Prevalence of AKs in SCC Specimens

The contribution of pre-existing AKs to the development of SCC has been assessed by examining invasive SCC specimens for contiguous or associated AKs. Studies have found that an AK was present in: - 60 percent of cases, - 82.4 percent, - 97 percent, - 97.2 percent, - and 100 percent of reviewed SCC specimens.

One prospective study found a contiguous AK within histopathologically confirmed SCC in 72 percent of cases, supporting the notion that most SCCs arise from precursor AK lesions.

Spontaneous Regression of AKs

Spontaneous regression of AKs is unpredictable for any individual lesion. One study reported that up to 25 percent of AKs regressed over a 1-year period, particularly with reduced sun exposure. Another study found higher regression rates in individuals who routinely used sunscreen.

AKs as Markers for Non-Melanoma Skin Cancer (NMSC)

The presence of AKs indicates chronic sun damage and serves as a clinical marker for increased risk of developing NMSC. Individuals with AKs are at elevated risk for SCC and basal cell carcinoma (BCC), and to a lesser extent, melanoma.

Metastatic Potential of SCC

The estimated metastasis rate for actinically derived SCC ranges from 1 percent to 20 percent, depending largely on tumor depth and anatomic location. The lip, ear, and scalp are sites associated with the highest risk of metastasis.

Treatment

Rationale for Treatment

Because it is impossible to predict which AKs will persist, regress, or progress to SCC, treatment decisions can be challenging. Although some argue that the low overall transformation rate makes treatment unnecessary, most dermatologists recommend treating AKs to prevent potential progression to invasive SCC. Treatment is also indicated to relieve symptoms such as pain and pruritus in affected patients.

Challenges in Treatment Selection

There are no standardized guidelines for AK management due to variability in study design, outcome measures, and follow-up duration. Even clinical diagnosis by expert dermatologists can be inconsistent.

Ideal Treatment Characteristics

The ideal therapy for AKs would be: - 100 percent effective and durable, - Clear both clinical and subclinical lesions, - Affordable, - Easy to administer or self-apply, - Convenient with no discomfort or downtime, - And yield excellent cosmetic results.

No single treatment currently meets all these criteria. The choice of therapy depends on the clinician's expertise, patient-specific factors, and the number, size, and location of lesions.

Treatment Categories

Therapies for AKs are broadly categorized as: - **Lesion-targeted treatments** - **Field therapies**

(Refer to Figure 113-6 for a management algorithm: 5-FU = 5-fluorouracil; PDT = photodynamic therapy; UVR = ultraviolet radiation.)